SUR - SSOT - Inventory Availability and Management

Reference Article and background information is from Tim Klemen see: Inventory Availability for SUR

Overview

This document outlines how different repair service provider models operate within Asurion, including part ownership, inventory responsibilities, and system interactions.

Key Terms

- **AOP (Asurion Owned Parts)**
  Parts owned, procured, and distributed by Asurion.
- **NAOP (Non-Asurion Owned Parts)**
  Parts owned and managed by external providers or clients.

Repair Service Provider Models

1. UBreakIFix (UBIF) as a Repair Service Provider (NAOP)

UBIF operates as a service provider in the US with integrated appointment booking and parts-driven availability.

Appointment Availability Logic:

- If **required part is in stock at store**:
  - Appointments are available based on technician capacity
- If **part is not in stock**:
  - Appointment window may be delayed (**Just-in-Time fulfillment**), or
  - Store may not appear as an option (if unavailable at UBIF distribution)

Financial & System Flow:

- Repairs are processed through **ServiceBench**
- ServiceBench determines the **authorized payment amount**
- Payment flow:
  - Asurion → UBIF Franchise Co.
  - UBIF Franchise Co. → Individual stores

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2. Asurion 3rd Party Repair Network (AOP)

Asurion partners with third-party repair providers to fill gaps in UBIF coverage in the US region

Responsibilities & Processes:

- **Parts Management**
  - Asurion WLI procures and distributes parts.
  - Parts are shipped via **Transfer Orders (TOs)**.
  - Providers must **receive TOs** to acknowledge inventory.
- **Inventory Management**
  - Providers are required to:
    - Perform **periodic inventory counts**
    - Submit **on-demand replenishment orders**
- **Returns Process**
  - When required, providers return inventory via **Return Transfer Orders (RTOs)**:
    1. Pick parts
    2. Package items
    3. Print shipping label
    4. Hand off to carrier (UPS, FedEx, etc.)

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3. Non-US Client-Specific 3rd Party Repair Network (NAOP)

Certain clients (e.g., Telus, ATT) operate their own repair networks using third-party providers.

Key Characteristics:

- Providers use **Non-Asurion Owned Parts (NAOP)**
- Asurion does **not manage inventory or distribution**
- Providers **expose part availability and orderability data** for their SKUs

Examples:

- Region: Canada → Client: Telus → Provider: Mobile Klinik
  - Note: Telus is moving to a new vendor in 2026 giving us an opportunity to change the integration from RTI to Tyk
- Region: Mexico → Client: ATT MXO → Regional third-party providers
  - Migrating from RTI to Tyk requires conversations

High-Level Scope of Features Impacted by this Project

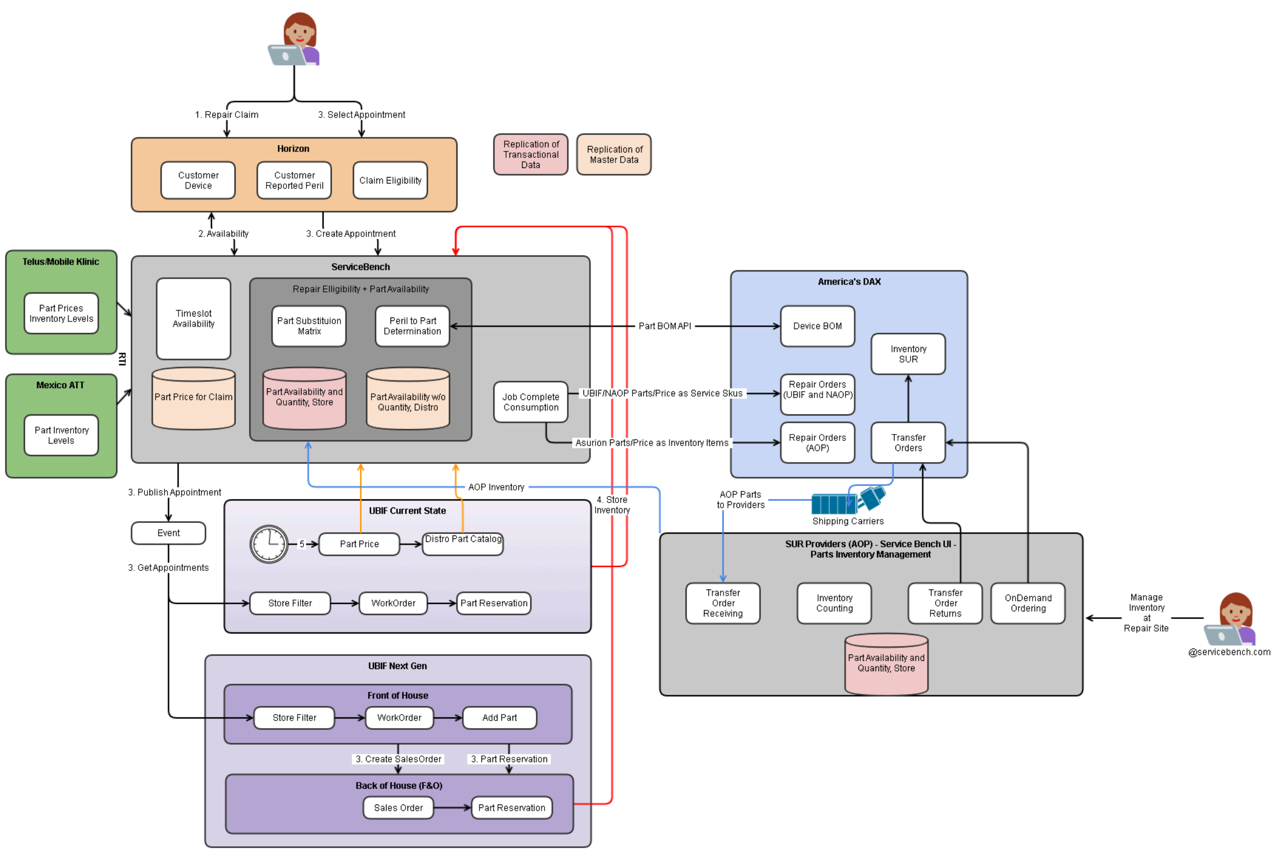
This document focuses on evaluating current and future architecture and system capabilities in relation to ServiceBench/Prism and D365 F&O, specifically:

	Capability	Current State
1	Peril to Part Type Configuration (e.g., CrackedFrontGlass → FrontGlass)	Managed in ServiceBench
2	Part Master Management • Part to Job Type Exclusions by Client • Part JIT Lead Time in Days • Part Replenishment Levels • Part Substitution Matrix by Client	Managed in ServiceBench

3	Inventory management User interfaces for Asurion Owned Parts(AOP) • Transfer Order Receiving and Returns • Inventory Counting • On-Demand Ordering - Creates a transfer order with Asurion WLI distribution warehouse	Managed in ServiceBench User Interface Access controlled via Service Bench user identity management Note: These inventory management activities such as receiving and counting write qtys to Service Bench database to be used during Availability queries.
4	Searching for Part Availability By (Repair Provider Location Level)	Stored in ServiceBench Database Uses direct database queries to check inventory for nearest geo located repair providers in a single query. No reservation concept exists, but a “min threshold buffer” qty is added to combat low stock scenarios
5	Part list enabled for Just-In-Time(JIT) (Distribution Level) • Americas WLI JIT Part List • UBIF Distro JIT Part List	Managed in ServiceBench by Users providing lists of Parts from both WLI and Distro that are allowed for JIT No qty levels maintained. Just 1 or 0. This combined with JIT lead time in days protects against stock shortages at the distribution centers.
6	Part Pricing for Claims	Managed in ServiceBench Some prices are sourced in Data Feeds (ie. Telus, Mexico*) UBIF recently migrated to communicating price in repair complete messaging. AOP uses zero prices as these parts are owned by Asurion and simply consigned to the AOP providers
7	UBIF Inventory Data Feeds • JSON SNS Events	Sent by UBIF Legacy and Next Gen Systems Consumed and Stored by Service Bench

8	Canada Telus/Mobile Klinic Inventory Data Feeds • RTI Integration	Sent by Client's repair vendor software system via ServiceBench's RTI(Real-Time Integration) apis. Consumed and Stored by Service Bench
9	Mexico ATT Inventory Data Feeds • RTI Integration	Sent by Client's repair vendor software system via ServiceBench's RTI(Real-Time Integration) apis. Consumed and Stored by Service Bench

Current Architecture



Data Replication

The current solution replicates three datasets from UBIF into ServiceBench to support the three features identified in Scope. ServiceBench uses this data locally during to support SUR

Availability, Just in Time, and Claims.

System Integration

In the current approach, replication of data between UBIF and ServiceBench is achieved via several integration patterns. Current State UBIF uses both events and APIs to update ServiceBench. Next Gen UBIF publishes events for ServiceBench and does not use API integration for these capabilities.

Risks of Data Replication

Replication of data from UBIF to ServiceBench has certain associated risks, regardless of the method of system integration.

- Data can become inconsistent across the two systems
- Data latency can occur, resulting in accurate but delayed data
- Conflicts can occur if both systems are acting on the data
- Scalability may require additional infrastructure (retries, idempotency, reconciliation)
- Risks are increased for high frequency, transactional data

Impacts of Failed Data Replication

The current solution replicates three datasets from UBIF into ServiceBench to support the three features identified in Scope. ServiceBench uses this data locally during to support SUR Availability, Just in Time, and Claims. In all three capabilities, there are risks if that data is not represented correctly.

Capability	Dataset	Notes	Transactional / Master	Impacts
Part Availability in Store/Provider location	Part Quantity by Store	• Current State and Next Gen UBIF send to ServiceBench separately • Various integration	Transactional	Inaccurate representation of Inventory can result in: • over - store cannot honor customer

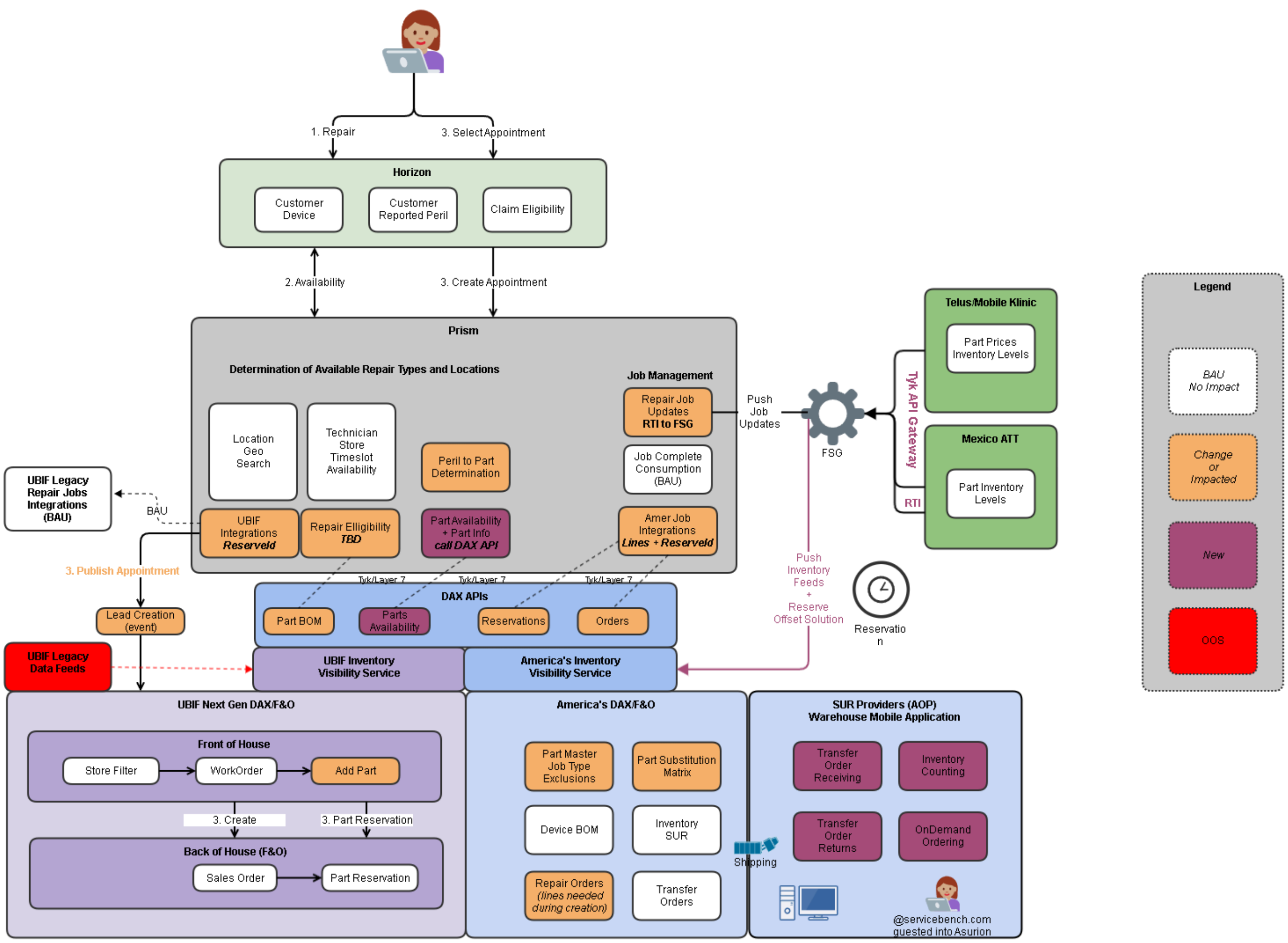
		patterns including event publication (UBIF Next Gen) and API • Transactional, changes frequently		appointment, replacement at repair cost • under / missing - missed revenue opportunity for a store, replacement instead of a repair
Just In Time AOP vs UBIF	Distro Part Catalog Americas WLI Part Catalog	• Distro Parts are available for Just in Time scheduling • Master, changes infrequently	Master	Inaccurate representation of Just in Time Parts can lead result in: • incorrectly included - store cannot honor customer appointment, replacement at repair cost • incorrectly missing - missed revenue opportunity for a store, replacement instead of a repair
Claims	Part Price	• Financial reconciliation between	Master	Inaccurate or missing part price can create

		Asurion and UBIF Franchise Co. • Master, changes infrequently		inaccurate results during financial reconciliation between UBIF Franchise Co. and Asurion.

Future Architecture

Conceptual Design

SUR - SSOT - Inventory Availability Orchestration



Scope

The Future Architecture can address one or all of the at risk capabilities indicated above. The replication of transactional data (part availability and quantity by store) is most significant and should be included in the scope of any changes.

Capability	Current State	Future State
Peril to Part Mapping (e.g., CrackedFrontGlass → FrontGlass)	Managed in ServiceBench	Remains in Prism/ServiceBench
Part Master Management • Part to Job Type Exclusions • Part JIT Lead Time in Days • Part Replenishment Levels • Part Substitution Matrix	Managed in ServiceBench	Data Ownership and Management Migrated to D365 F&O • **Any "repair eligibility/decision impacting" business logic on these data points will be handled exclusively by Prism**
Inventory management User interfaces for Asurion Owned Parts(AOP) • Transfer Order Receiving and Returns • Inventory Counting • On-Demand Ordering - Creates a transfer order with Asurion WLI distribution warehouse	Managed in ServiceBench User Interface Access controlled via Service Bench user identity management Note: These inventory management activities such as receiving and counting write qtys to Service Bench database to be used during Availability queries.	Migrated to D365 F&O Warehouse Mobile Application User Access controlled via D365 F&O with Azure Identity management Requires guesting of Service Bench user into Azure tenant.
Searching for Part Availability By (Repair Provider Location Level)	Managed in ServiceBench	Migrated to D365 F&O • A new Parts Inventory API will be provided so Prism can perform the inventory

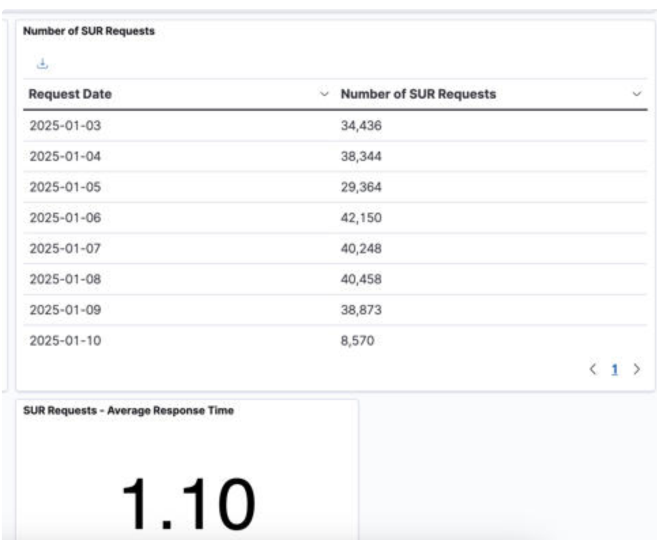
		check across multiple locations spanning US AOP, Canada, Mexico, and US UBIF
Part list enabled for Just-In-Time(JIT) (Distribution Level) • Americas WLI JIT Part List • UBIF Distro JIT Part List	Managed in ServiceBench	Migrated to D365 F&O • A new Parts Inventory API will be provided so Prism can perform the inventory check and in that reply see if the Part is enabled for just in time. Note: This only applies for US AOP and US UBIF repair locations. • Recommend WLI and Distro JIT enable flag be added to existing Americas Product Part Master
Part Pricing for Claims	Managed in ServiceBench Some prices are sourced in Data Feeds (ie. Telus, Mexico*) UBIF recently migrated to communicating price in repair complete messaging. AOP uses zero prices as these parts are owned by Asurion and simply consigned to the AOP providers	UBIF will send price in repair complete communications Prism will send final price in consumption message This has already been achieved and is live in Production. **Open Item: What about Non-UBIF NAOP and AOP prices?** Answer: Need more discussion for Mobile Klinic price management at sku level. Mexico needs more research on how they send prices during consumption today and if we have to solve for anything or if its BAU?

UBIF Inventory Data Feeds • JSON SNS Events	Sent by UBIF Legacy and Next Gen Systems Consumed and Stored by Service Bench	UBIF Next Gen will automatically feed into the UBIF inventory visibility service and be consumable by our DAX API layer. UBIF Legacy is in run off and is rapidly migrating to Next Gen. We will be evaluating if it's worth investing any development in solving for UBIF legacy. If so, our current plan would be to ingest the existing json data feed into our inventory visibility service.
Canada Telus/Mobile Klinic Inventory Data Feeds • RTI Integration	Sent by Client's repair vendor software system via ServiceBench's RTI(Real-Time Integration) apis. Consumed and Stored by Service Bench	**Tyk → FSG → DAX API*** Mobile Klinic is moving to a new CRM and has an opportunity to migrate from the legacy RTI interface to the more modern Tyk → FSG interface. **This Tyk/FSG migration will be handled by the Service Bench team**. **And Service Bench Team will route the data to DAX API for inventory visibility service ingestion.**
Mexico ATT Inventory Data Feeds • RTI Integration	Sent by Client's repair vendor software system via ServiceBench's RTI(Real-Time Integration) apis.	**Preferred but will take time: Tyk → FSG → DAX API*** Mexico has 4 different repair vendors and some are open to

	Consumed and Stored by Service Bench	change while others do not have capacity to change. **Any future Tyk/FSG migration where possible will be handled by the Service Bench team.** **And Service Bench Team will route the data to DAX API for inventory visibility service ingestion.**

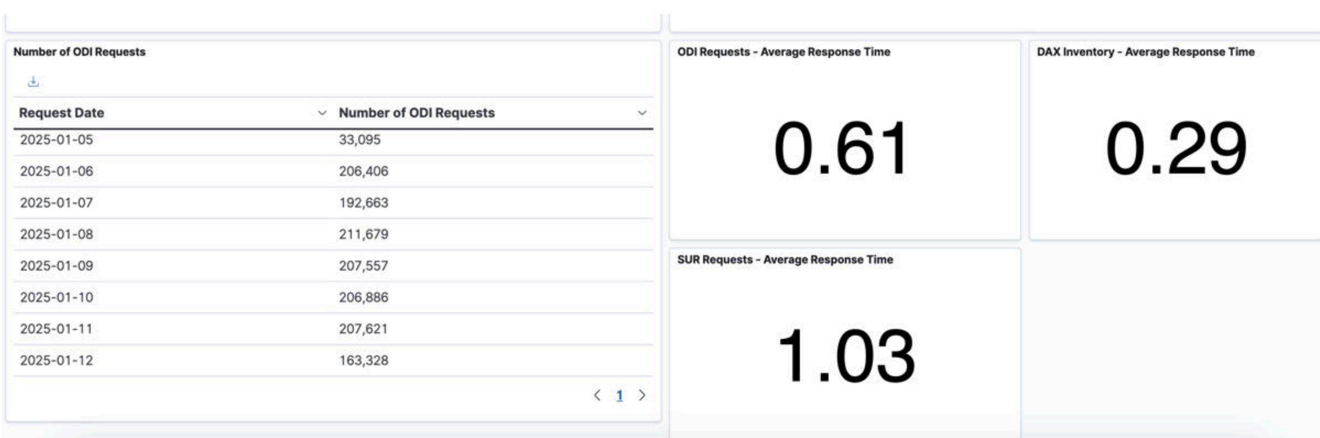
References

Servicebench Performance



Number of SUR Requests

Request Date	Number of SUR Requests
2025-01-03	34,436
2025-01-04	38,344
2025-01-05	29,364
2025-01-06	42,150
2025-01-07	40,248
2025-01-08	40,458
2025-01-09	38,873
2025-01-10	8,570



Number of ODI Requests

Request Date	Number of ODI Requests
2025-01-05	33,095
2025-01-06	206,406
2025-01-07	192,663
2025-01-08	211,679
2025-01-09	207,557
2025-01-10	206,886
2025-01-11	207,621
2025-01-12	163,328